

Flight Sim S.O.P. for Two-Stage Rockets

IREC Propulsion

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Purpose

The purpose of this document is to explain how to accurately run flight simulations on a two-stage rocket using OpenRocket and RASAero II.

Scope

This procedure covers the setup and execution of flight simulations for two-stage rockets using OpenRocket and RASAero II in tandem. The procedure begins with loading rocket (.ork and .CDX1) and motor files (.eng) into both flightsim software and ends with the export of flight simulation data from RASAero II for further analysis.

This procedure applies to any two-stage rocket configuration that needs to be simulated and assumes the user has access to finalized ork and CDX1 files as well as ENG files for both the booster and sustainer motors.

Equipment List

- OpenRocket software
- RASAero II software
- Engine files for both the booster motor and the sustainer motor (ENG files)
- An OpenRocket rocket file (ork file)
- A RASAero II rocket file (CDX1 file)

Procedure

1. Launch OpenRocket, then open the ork you want to run sims with by clicking [File] → [Open...]. After selecting the correct ork file, warnings may appear similar to what is shown below in Figure 1. If warnings appear, select [OK]. Your screen should look similar to what is shown below in Figure 2 if everything was done correctly.

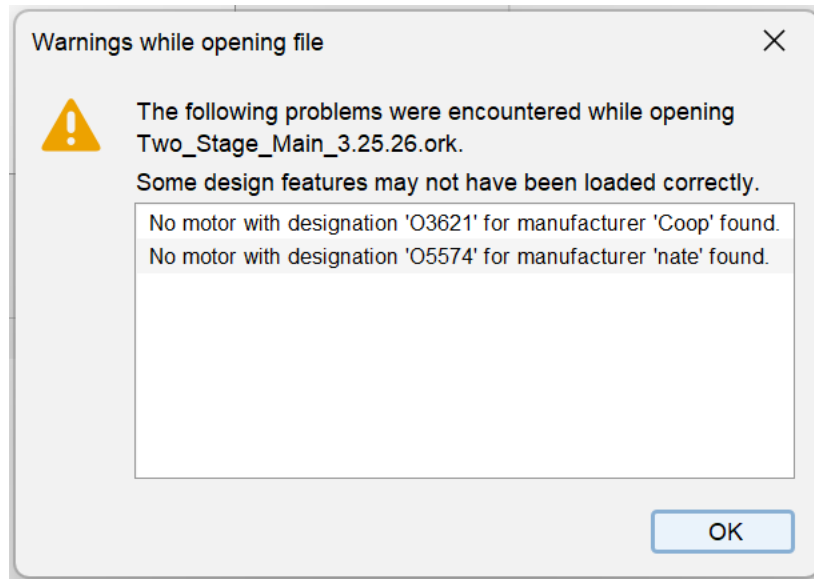


Figure 1: Warnings that may appear when opening an ork file.

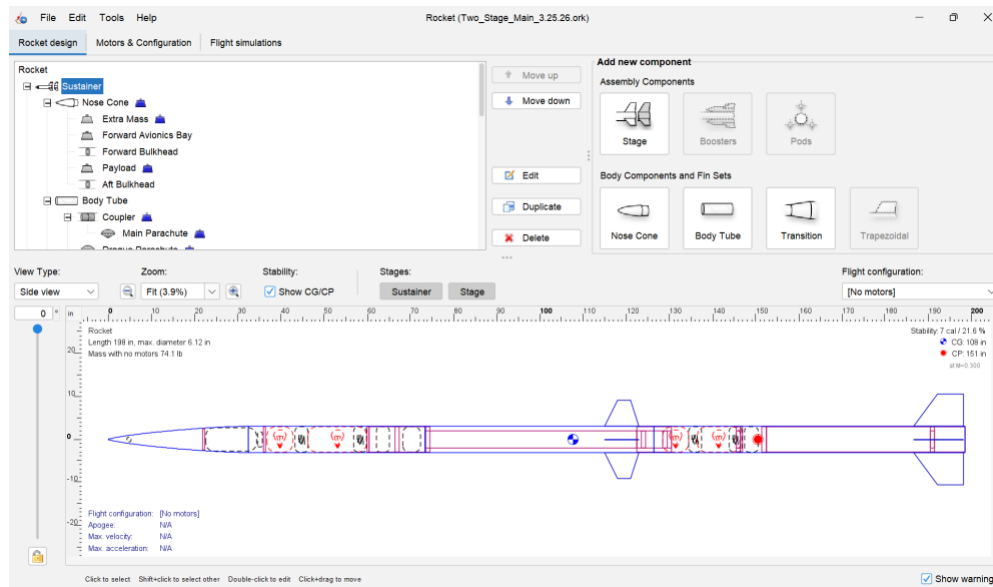


Figure 2: A successfully loaded ork file in OpenRocket.

- Next, ENG files for both the booster and the sustainer motor need to be uploaded. Select [Edit] → [Preferences]. Then, under where it says *User-defined thrust curves*, select [Add]. Then, select the directory that contains the ENG files you want to use and select [Add]. Your screen should now look similar to Figure 3 below. After the correct folder has been selected, select [OK].

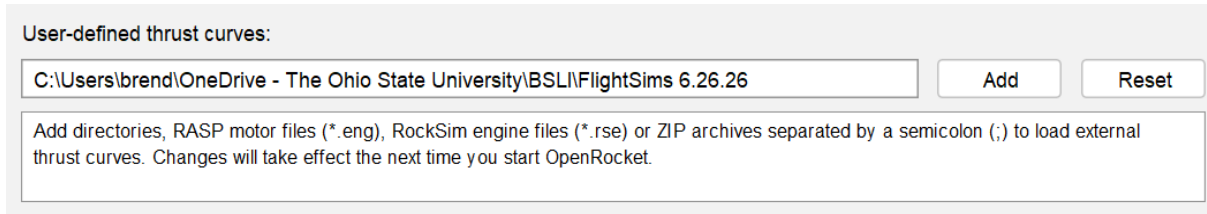


Figure 3: Options for importing user-defined thrust curves.

- Restart the OpenRocket software.** If you do not restart OpenRocket, you will not be able to find the ENG files you just added. After the software has been restarted, reopen the ork file by following the steps in Step 1. Next, select [Motors & Configuration] → [New Configuration]. Then, select the motor that you want to use for the booster. Ensure that the *Ejection charge delay* option is set to *Plugged*. After all correct options have been selected, press [OK]. If everything has been correctly selected, the *Motor configurations* tab should look similar to Figure 4.

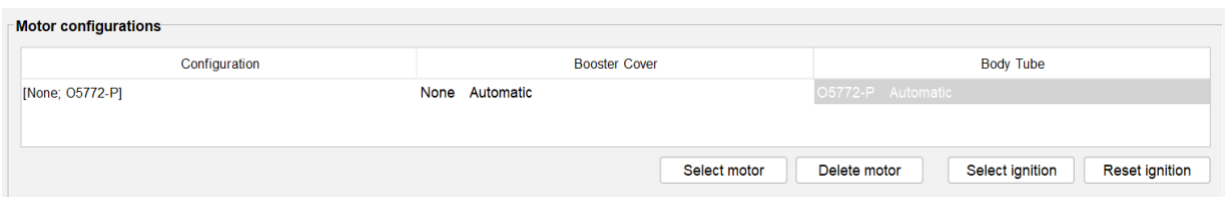


Figure 4: Motor configurations tab after selecting a booster motor.

- After the booster motor has been selected, the sustainer motor needs to be selected. Under the *Booster Cover* tab shown above in Figure 4, select where it says [None Automatic] so that is highlighted blue, then press [Select motor]. Next, select the motor that you want to use for your sustainer. Your screen should look similar to Figure 5 below if everything was done correctly.

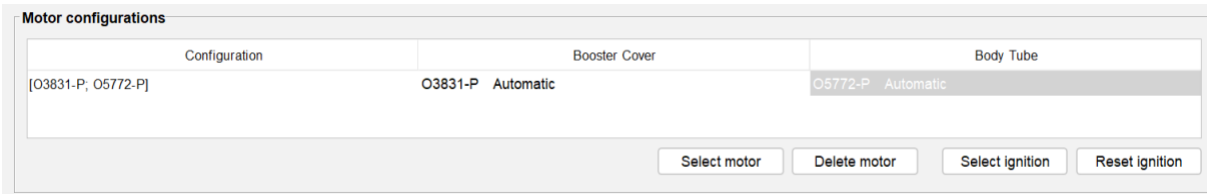


Figure 5: Motor configurations tab after selecting a booster and a sustainer motor.

- Keep your OpenRocket tab open and now open RASAero II. In RAS, select [File] → [Open], then select the CDX1 file you want to use and press [Open]. Next, select [File] → [Select Motor File], then select the same eng directory that you used in OpenRocket, then select [Open]. Your screen should look similar to Figure 6 below.

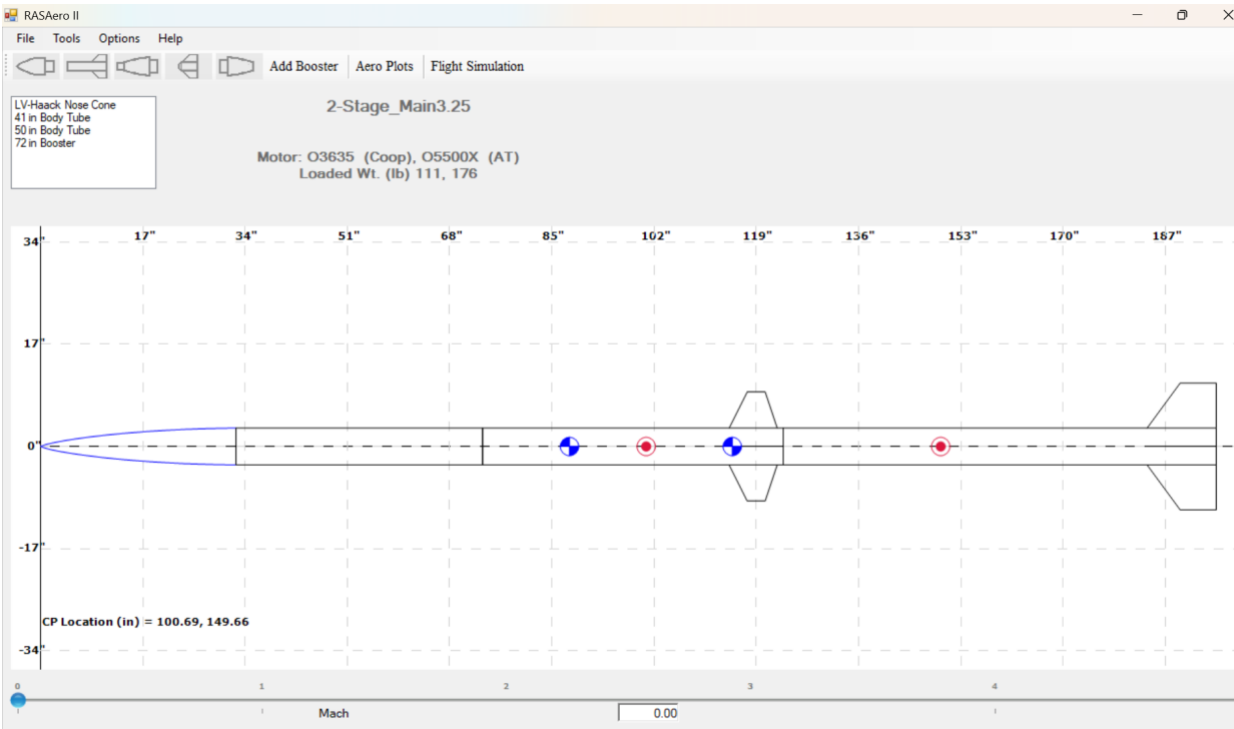


Figure 6: A successfully loaded rocket in RASAero II.

- Click on [Options] → [Launch Site], then fill in the correct information in each box. When done, select [OK]. Next, click on [Options] → [Surface Finish], and ensure that *Rough Camouflage Paint* is selected.

7. Select [Flight Simulation], then double click the box under the *Motor(s) Loaded* tab. Then, select or enter the correct information under every box that can be edited. Values for *Sustainer CG* and *Loaded Wt* can only be found in OpenRocket. Figures 7, 8, 9, and 10 show where values for *Sustainer CG*, *Sustainer Loaded Wt*, *Combined CG*, and *Combined Loaded Wt* can be found within OpenRocket. Ensure that the correct options boxed in red are selected for each value, then press [Save]. Your screen should be similar to what is shown in Figure 11 before saving.

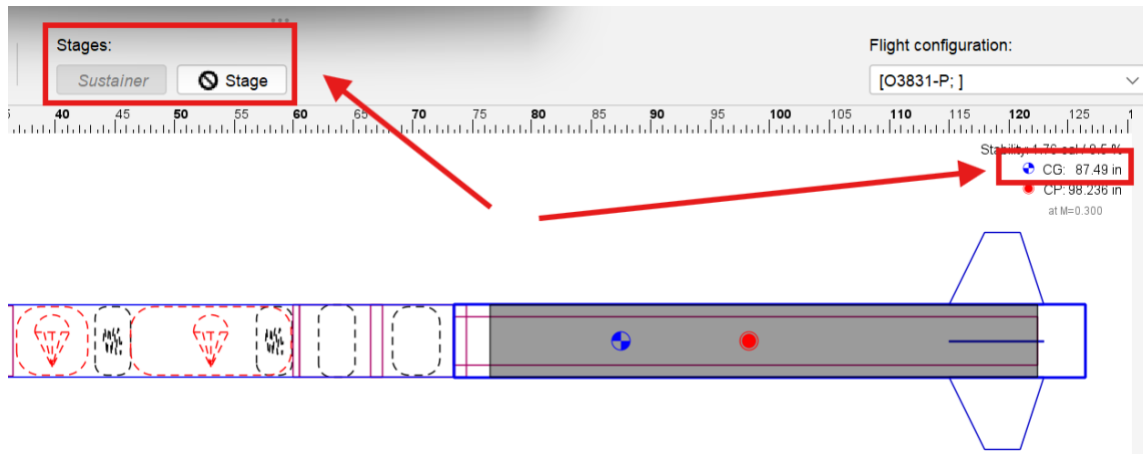


Figure 7: Location for finding Sustainer CG in OpenRocket.

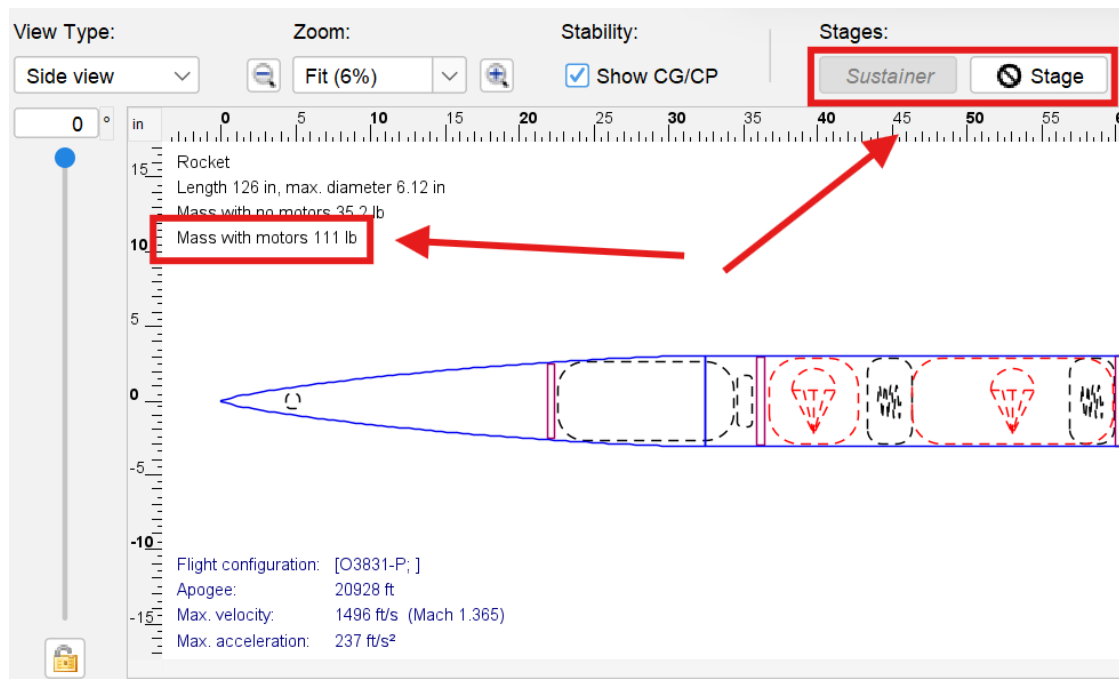


Figure 8: Location for finding Sustainer Loaded Wt in OpenRocket.

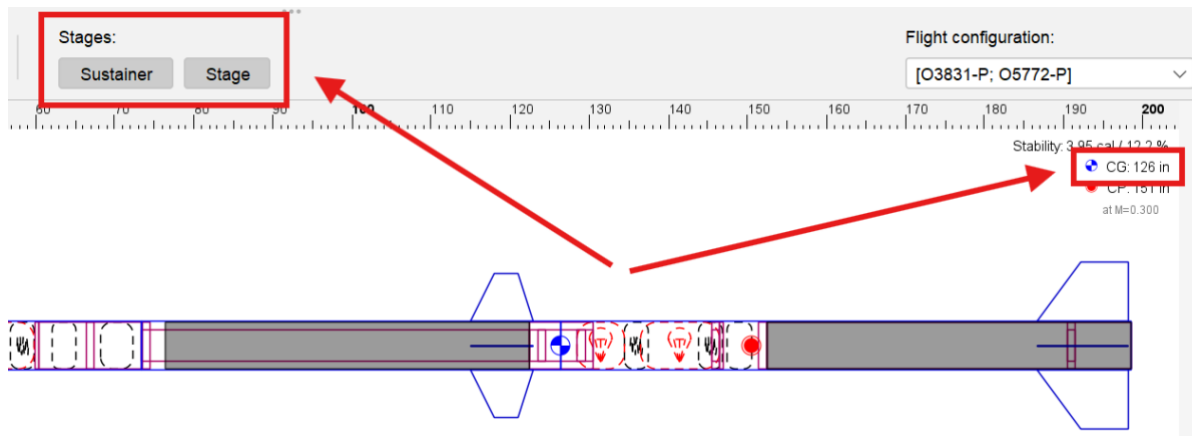


Figure 9: Location for finding Combined CG in OpenRocket

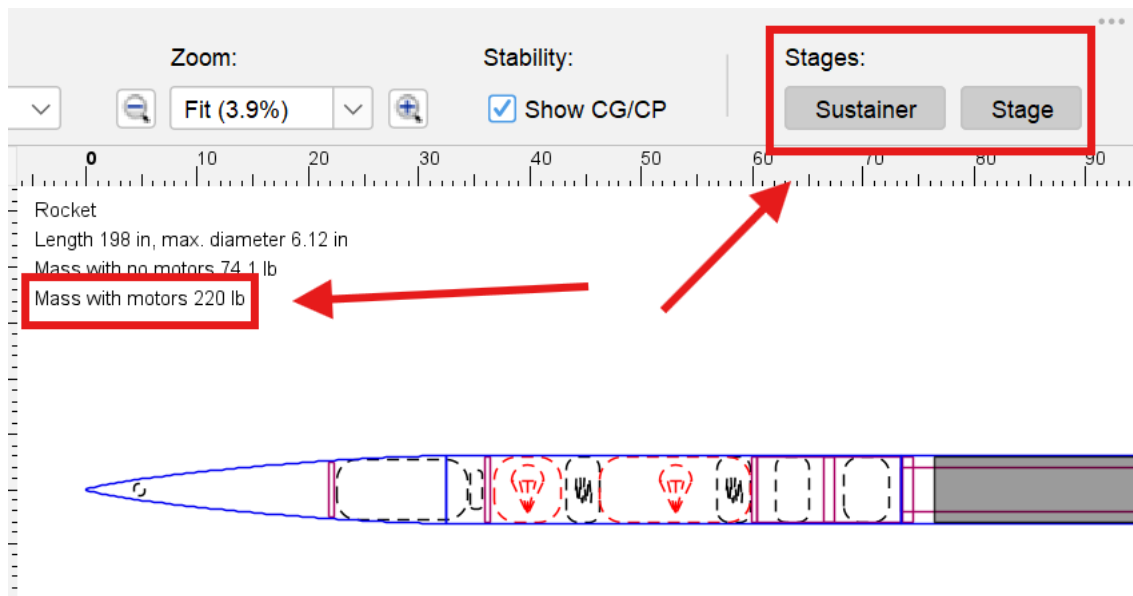


Figure 10: Location for finding Combined Loaded Wt in OpenRocket.

FlightDataEntry

Sustainer Motor

Nozzle Exit Diameter (in)

Sustainer CG (in)

Sustainer Loaded Wt (lb)

Ignition Delay (sec)

☒ Include Booster

Booster Motor

Nozzle Exit Diameter (in)

Separation Delay (sec)

Combined CG (in)

Total Loaded Wt (lb)

Ignition Delay (sec)

☐ Include Booster

Booster Motor

Nozzle Exit Diameter (in)

Separation Delay (sec)

Combined CG (in)

Total Loaded Wt (lb)

Save Cancel

Figure 11: Sample motor options in RASAero II.

8. Now, it is time to run the flight simulation. Under the *Flight simulations* tab, select [Simulations] → [Rerun All Simulations], then scroll to the right and press [View Data]. Data can then be exported from this screen and further analyzed. Example data is shown below in Figure 12.

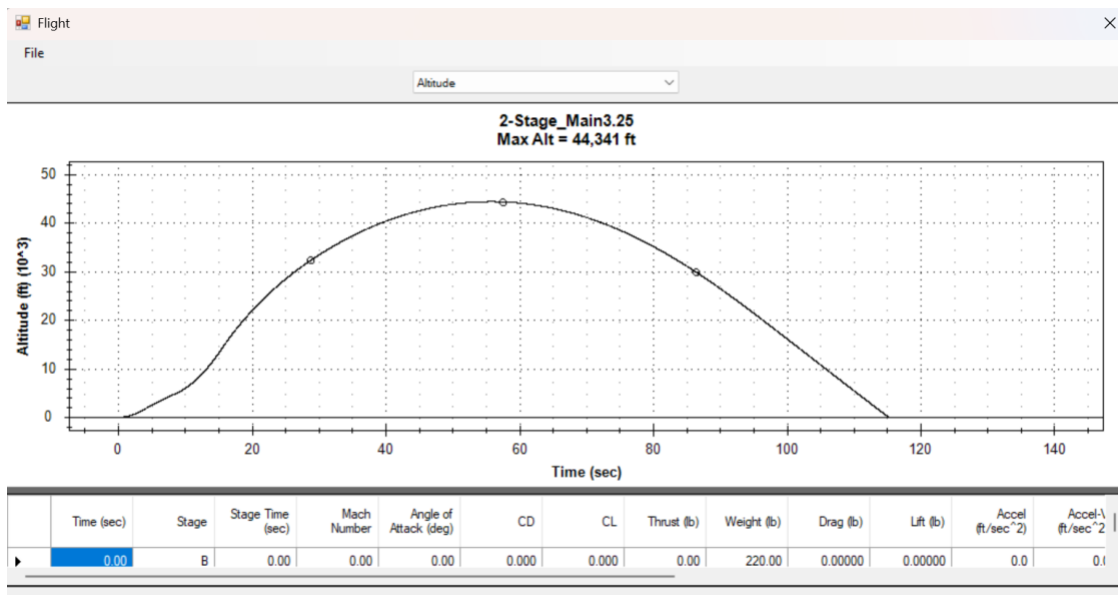


Figure 12: Sample flight simulation data.